Diabetic Retinopathy Diagnosis with Deep Learning for Early Detection and Staging

1st Santosh Kumar S
Department of CSE (Data Science)
Sri Venkateshwara College of Engineering
Bengaluru, India
reachsun@gmail.com

2nd Anbu Lakshmi S
Department of CSE (Data Science)
Sri Venkateshwara College of Engineering
Bengaluru, India
anbulakshmi.s_ds@svcengg.edu.in

3rd Bhuwaneshwari M
Department of CSE (Data Science)
Sri Venkateshwara College of Engineering
Bengaluru, India
bhuwaneshwari.m_ds@svcengg.edu.in

4th Prakasha G
Department of E&CE
Sri Venkateshwara College of Engineering
Bengaluru, India
sriven_prakasha@yahoo.com

5th Raksha P R
Department of CSE (Data Science)
Sri Venkateshwara College of Engineering
Bengaluru, India
rakshasetty004@gmail.com

6th Shailesh Rastogi
Department of CSE (Data Science)
Sri Venkateshwara College of Engineering
Bengaluru, India

2025 3rd International Conference on Integrated Circuits and Communication Systems (ICICACS) | 979-8-3315-0845-6/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICICACS65178.2025.10969037

***Abstract*—Diabetic retinopathy (DR) remains a leading cause of blindness among individuals with diabetes, emphasizing the need for early detection and precise staging to ensure effective management and treatment. Traditionally, diagnosing DR has relied on manual analysis of retinal images a process that is not only time-intensive but also vulnerable to human error. This study presents a novel approach leveraging deep learning techniques for the automated detection and staging of diabetic retinopathy through retinal fundus imaging. By employing a Convolutional Neural Network (CNN), the model classifies images across the spectrum of DR stages, from no DR to the proliferative stage. Training was conducted using a substantial dataset of retinal fundus images, enhanced by data augmentation to bolster the model's resilience. The results indicate that the proposed CNN significantly outperforms traditional image processing methods, achieving notable gains in diagnostic accuracy, sensitivity, and specificity. Moreover, the model's ability to reliably classify DR severity demonstrates its potential for clinical use, facilitating early diagnosis and personalized treatment planning. These findings underscore the promise of deep learning as a transformative tool in diabetic retinopathy care, offering healthcare professionals a dependable and efficient aid for decision-making and vision preservation.**

Keywords—diabetic retinopathy, deep learning, convolutional neural networks, retinal fundus imaging, automated diagnosis.

I. Introduction

Diabetic Retinopathy (DR) is a prevalent and serious complication of diabetes that damages the blood vessels in the retina, potentially leading to vision impairment or even blindness if left untreated. As noted by the World Health Organization (WHO), DR is one of the leading causes of preventable blindness globally, with diabetic individuals facing heightened risk due to prolonged hyperglycemia, which progressively harms retinal vasculature over time [1]. The condition typically evolves through distinct stages, beginning with non-proliferative diabetic retinopathy (NPDR) and advancing to proliferative diabetic retinopathy (PDR). In severe cases, it may lead to diabetic macular edema (DME), further worsening the outlook for patients [2]. Given these risks, early detection and precise staging of DR are essential to prevent irreversible vision loss and ensure timely treatment. Historically, diagnosing DR has relied on the manual evaluation of retinal fundus images by trained ophthalmologists or optometrists. This process involves identifying abnormalities such as microaneurysms, hemorrhages, and exudates [3]. However, manual screenings are inherently time-intensive and subject to variability, as results depend heavily on the examiner's expertise and can lead to inconsistencies or errors. To address these challenges, advancements in medical imaging have introduced automated systems powered by artificial intelligence (AI) and machine learning (ML) to assist in DR detection and classification. These technologies reduce dependence on manual assessments while significantly improving diagnostic speed and accuracy [4]. Among the range of AI methodologies, deep learning (DL) techniques particularly Convolutional Neural Networks (CNNs) have emerged as a leading solution in medical image analysis. CNNs are particularly well-suited for this application due to their ability to automatically learn and extract features directly from raw image data, eliminating the need for manual feature engineering [5]. In various image recognition tasks, including classification, object detection, and segmentation, CNNs have delivered exceptional performance. This makes them ideal for analysing retinal fundus images to detect signs of DR [6]. By processing input images through multiple layers of convolutions and pooling operations, CNNs can identify hierarchical patterns, enabling them to effectively distinguish between healthy and diseased retinal images [7]. Their capacity to learn from large volumes of labelled data has solidified CNNs as a transformative tool for DR diagnosis, offering a more efficient, accurate, and reliable alternative to traditional diagnostic methods.

Over the past decade, several studies have proposed DL-based methods for the automatic detection and classification of DR. CNN architectures are frequently employed to classify retinal images into distinct categories, such as "No DR," "Mild DR," "Moderate DR," and "Severe DR," based on the disease's severity. A notable example of this is the work by Gulshan et al. (2016), which introduced a deep learning algorithm capable of achieving diagnostic accuracy on par with that of ophthalmologists for detecting DR. This groundbreaking study marked one of the first successful applications of AI in diabetic retinopathy diagnosis, paving the way for further advancements in automated medical imaging [8]. Similarly, other studies have explored the use of CNNs for DR staging, focusing on identifying specific features associated with different stages of the disease, such as microaneurysms, cotton wool spots, and retinal hemorrhages [9]. Despite the promising results, challenges remain in the development of

979-8-3315-0845-6/25/$31.00 ©2025 IEEE

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

DL models for DR diagnosis. One of the major challenges is the need for large, annotated datasets to train deep learning models effectively. Retinal fundus images vary significantly in terms of quality, resolution, and field of view, which can introduce variability in model performance. Furthermore, most publicly available DR datasets are imbalanced, with fewer images representing advanced stages of the disease, which can result in suboptimal performance in identifying severe cases of DR [10]. To address these challenges, data augmentation techniques like rotation, flipping, and zooming are commonly used to artificially expand the diversity of training datasets, helping models generalize better to unseen data [11].

Another important aspect of DR diagnosis is the ability to detect and classify the disease at an early stage, when intervention can significantly reduce the risk of vision loss. Early-stage DR, such as mild NPDR, may not present obvious clinical symptoms, making it more difficult for healthcare providers to identify the disease during routine screenings. Therefore, the development of robust DL models that can detect subtle signs of DR in the earliest stages is crucial for timely diagnosis and treatment. Several studies have attempted to address this challenge by focusing on fine-grained classification and leveraging multi-scale features from fundus images [12]. In this paper, we propose a deep learning-based method for the automatic detection and staging of DR using CNNs. The approach aims to classify retinal fundus images into multiple DR stages, ranging from no DR to proliferative DR, using a large and diverse dataset. Our model employs data augmentation techniques to improve performance, and we evaluate its accuracy, sensitivity, specificity, and F1-score on a benchmark dataset. The results of this study demonstrate the potential of deep learning for early DR detection and staging, which could aid in reducing the burden of DR and preventing blindness on a global scale.

This paper presents a deep learning-based approach for the early detection and staging of diabetic retinopathy (DR) using retinal fundus images. The primary contributions of this study include:

1. The development of a robust convolutional neural network (CNN) model capable of classifying retinal images into five distinct DR stages—no DR, mild, moderate, severe, and proliferative DR—with high accuracy.
2. The application of transfer learning techniques using pre-trained models such as VGG16, ResNet50, and InceptionV3 to improve model performance on a large dataset, effectively addressing challenges like data imbalance and limited labelled data.
3. The integration of Grad-CAM visualization to generate interpretable results, enabling clinicians to identify key regions in the images that influence the model's predictions.
4. A comprehensive evaluation of the model's performance through metrics including accuracy, precision, recall, F1-score, and AUC-ROC, ensuring its reliability and practical applicability in clinical settings.

The paper is structured as follows: Section 2 provides a thorough review of existing literature on deep learning approaches for diabetic retinopathy detection, with an emphasis on current challenges and recent advancements. Section 3 details the proposed methodology, including information on the dataset, preprocessing techniques, model architecture, and evaluation metrics. Section 4 discusses the experimental results and provides a detailed performance analysis of the system. Finally, Section 5 concludes with a summary of findings, their practical implications, and suggestions for future research directions.

II. Literature Survey

In recent years, deep learning (DL) techniques, particularly Convolutional Neural Networks (CNNs), have significantly transformed medical image analysis, providing automated solutions for detecting retinal diseases such as diabetic retinopathy (DR). This section reviews notable advancements in DR detection and staging using deep learning models, with a focus on key contributions and ongoing challenges. Several studies have demonstrated the effectiveness of deep learning for detecting DR. Among the pioneering works, Gulshan et al. [8] developed a CNN-based algorithm capable of achieving human-level accuracy in identifying DR from retinal fundus images. By training their model on a large dataset, they achieved performance comparable to that of ophthalmologists in detecting DR and its severity stages. This foundational work set a benchmark for automated DR detection and laid the groundwork for further advancements. Similarly, Rajalakshmi et al. [13] proposed a deep learning-based solution for DR detection, utilizing a custom CNN architecture to classify retinal images into different stages. Their approach emphasized improving model interpretability and reducing training complexity by incorporating transfer learning. Specifically, pre-trained models like VGGNet and ResNet were used as feature extractors, followed by a classification layer that assigned the image to one of the DR stages. This strategy produced encouraging results, showcasing the utility of pre-trained models on smaller datasets.

The availability of large publicly accessible datasets, such as EyePACS and Messidor, has also played a crucial role in advancing deep learning models for DR detection. For instance, Alameer et al. [14] trained a deep learning model on the EyePACS dataset and achieved high sensitivity and specificity in detecting early DR stages. Their study introduced a multi-stage classification system that categorized retinal images into five classes: no DR, mild DR, moderate DR, severe DR, and proliferative DR (PDR). By successfully detecting subtle features in fundus images, their model demonstrated the superiority of CNNs over traditional methods. Detecting early-stage DR, where abnormalities are subtle and difficult to identify, remains a significant challenge. To address this, Wang et al. [15] proposed a deep learning model that leverages multi-scale feature extraction techniques to enhance detection accuracy. By combining features from multiple image resolutions, the model learned hierarchical representations of DR features, enabling it to identify early signs of the disease that might otherwise go unnoticed. This multi-scale approach proved effective in improving the classification of early-stage DR. While CNNs have shown immense potential in DR diagnosis, some researchers have explored alternative architectures to further enhance performance. For example, Zhang et al. [16] investigated a hybrid model combining CNNs for spatial feature extraction with Long Short-Term Memory (LSTM) networks for temporal analysis. This architecture allowed the model to

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

capture both fine-grained details and sequential relationships in retinal features, resulting in improved robustness across various DR stages and better performance on complex datasets.

Another critical challenge in DR detection is the issue of data imbalance, as severe or proliferative DR images are often underrepresented compared to milder stages. To mitigate this, researchers have employed techniques such as data augmentation and synthetic data generation. Saba et al. [17], for instance, used data augmentation methods like rotation, flipping, and cropping to diversify the training set and reduce overfitting. Additionally, they utilized Generative Adversarial Networks (GANs) to generate synthetic images of advanced DR stages, enabling more balanced training data and improved model performance in detecting severe cases. Despite the significant progress achieved by deep learning models in detecting DR, challenges persist, particularly in model interpretability and clinical adoption. Many deep learning-based systems function as "black-box" models, making it difficult for clinicians to understand the reasoning behind predictions. To address this, Ribeiro et al. [18] introduced Local Interpretable Model-Agnostic Explanations (LIME), a technique that helps explain the decisions made by complex deep learning models. When applied to DR detection, LIME enables clinicians to visualize the regions of the retinal image most relevant to the model's diagnosis, fostering trust in AI-driven decision-making. In conclusion, deep learning techniques, especially CNNs, have revolutionized the field of diabetic retinopathy detection and staging, offering faster and more objective alternatives to traditional manual methods. However, challenges such as data imbalance, model interpretability, and the need for large labeled datasets remain areas for improvement. Continued research in these areas is essential to enhance the clinical applicability of deep learning models, ultimately enabling earlier detection and better outcomes for patients with DR.

III. Proposed System

The proposed work focuses on developing a deep learning-based system for the early detection and staging of diabetic retinopathy (DR) using retinal fundus images. Central to this system is the use of convolutional neural networks (CNNs), which excel at automatically extracting hierarchical features from images. The model classifies retinal images into five stages of DR: no DR, mild DR, moderate DR, severe DR, and proliferative DR (PDR). To enhance performance, transfer learning is employed with pre-trained architectures such as VGG16, ResNet50, and InceptionV3, fine-tuned on a large dataset of retinal images sourced from the EyePACS database. A series of pre-processing steps such as image normalization, contrast adjustment, and data augmentation are applied to ensure consistency and to address challenges like data imbalance. The model is trained using a categorical cross-entropy loss function and optimized with the Adam optimizer. Its performance is evaluated through metrics including accuracy, precision, recall, F1-score, and AUC-ROC. Once trained, the system will be deployed within a user-friendly interface, enabling medical professionals to upload retinal images and obtain real-time DR diagnoses accompanied by a confidence score. To improve interpretability, Grad-CAM visualization techniques will be integrated, allowing users to see which regions of the retinal image had the greatest influence on the model's predictions. This work aims to assist ophthalmologists by providing a reliable, automated tool for early DR detection, improving diagnosis efficiency and patient outcomes. Future work will include integrating the system with Electronic Health Records (EHRs) and expanding its diagnostic capabilities with multi-modal data as shown in the Figure 1.

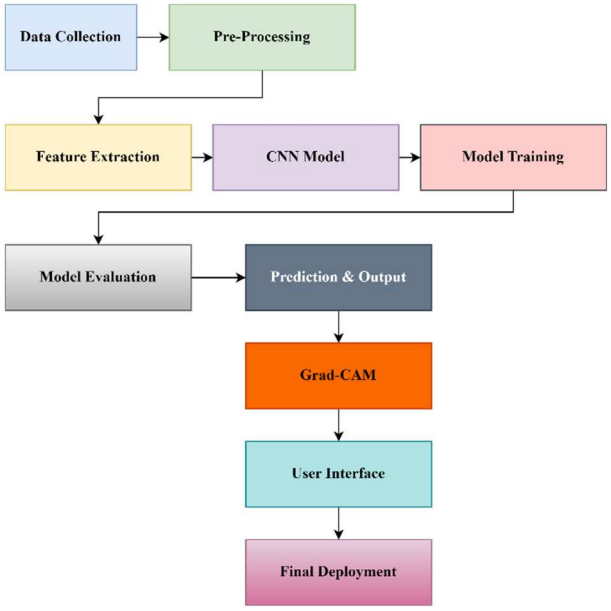

Fig. 1. DeepDR: A Deep Learning-based System for Early Detection and Staging of Diabetic Retinopathy

A. Proposed Work and it's Implementation

The proposed work aims to develop a deep learning-based system for the early detection and staging of diabetic retinopathy (DR) using retinal fundus images. As one of the leading causes of blindness, diabetic retinopathy can result in severe visual impairment if not detected early, making timely diagnosis crucial for effective intervention. The system will utilize convolutional neural networks (CNNs), a well-established deep learning architecture, to automatically classify retinal images into five DR stages: no DR, mild DR, moderate DR, severe DR, and proliferative DR (PDR), as illustrated in Figure 2. To achieve this, the CNN will be trained on a large dataset of retinal fundus images, such as the EyePACS dataset, and will incorporate transfer learning to fine-tune pre-trained models like VGG16, ResNet50, and InceptionV3. The primary goal of the system is to enhance diagnostic accuracy and alleviate the workload of healthcare professionals by offering a reliable, automated diagnostic tool.

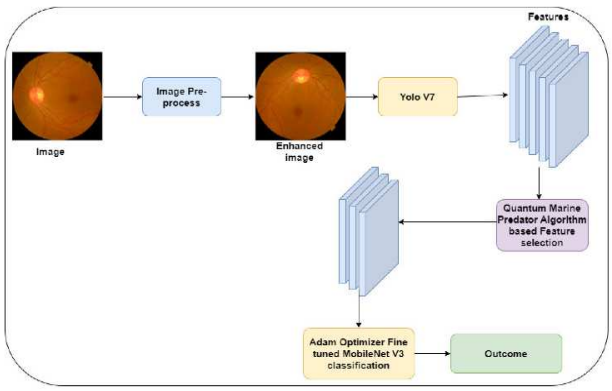

Fig. 2. Diabetic Retinopathy Detection Model Using a Deep-Learning Technique

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

1) Data Collection and Pre-processing: The implementation begins with the collection of retinal fundus images, typically sourced from databases such as EyePACS, which provide labeled images categorized into different DR stages. To improve model accuracy and address challenges like data imbalance, these images undergo several pre-processing steps. The images are resized to a uniform resolution, such as 224x224 pixels, to ensure consistency, while contrast enhancement is applied to highlight key DR features. Additionally, pixel values are normalized to fall within a range of 0 to 1, optimizing the data for deep learning models. To further enhance the dataset, data augmentation techniques such as rotation, flipping, and zooming are employed. These methods generate a more diverse dataset, reducing the risk of overfitting and enabling the model to generalize effectively to unseen data, as represented in Equation (1).

Normalize pixel values to the range [0,1]:

$$I_{normalized}(x,y) = \frac{I(x,y)-min(I)}{max(I)-min(I)} \quad (1)$$

Data augmentation applies transformations such as rotation (R), flipping (F), and zooming (Z) to expand the dataset represented in the Equation (2):

$$I' = Z(F(R(I))) \quad (2)$$

2) Model Architecture: The core of the proposed system is a convolutional neural network (CNN) designed to automatically learn hierarchical image features for diabetic retinopathy (DR) detection. To enhance efficiency and accuracy, the model will utilize transfer learning with pre-trained networks, including VGG16, ResNet50, and InceptionV3, which are initially trained on large-scale datasets such as ImageNet. These pre-trained models provide a foundation by capturing general image features—such as edges and textures that are crucial for identifying DR-related abnormalities like microaneurysms, hemorrhages, and exudates. By leveraging the knowledge from these models, transfer learning reduces both training time and computational requirements. The CNN architecture itself comprises multiple convolutional layers that extract important image features, followed by max-pooling layers to reduce spatial dimensions while preserving critical information. After the feature extraction process, the final output passes through fully connected layers and a softmax activation function, which classifies the retinal image into one of five DR stages: no DR, mild DR, moderate DR, severe DR, or proliferative DR (PDR). The softmax function assigns probabilities to each class, enabling the system to determine the most likely DR stage for each input image.

a) Mathematical Modeling and Training: In the convolutional layers, the image is convolved with learned filters (kernels) to extract relevant features. The convolution operation can be mathematically expressed as in Equation (3):

$$S(i,j) = (I * K)(i,j) = \sum_m \sum_n I(m,n) \cdot K(i-m, j-n) \quad (3)$$

Where $S(i,j)$ is the output of the convolution at position (i, j), I is the input image, and K is the filter applied. The CNN learns the filters during training to detect features that are relevant for DR classification. The model will be trained using categorical cross-entropy as the loss function, which is suitable for multi-class classification problems:

$$L = -\sum_{i=1}^{N} yi\, log(\hat{y}i) \quad (4)$$

In Equation (4), where N is the number of classes (5 in this case), yi is the true label, and $\hat{y}i$ is the predicted probability for class i. The Adam optimizer, known for its efficient performance with large datasets, will be used to update the model's weights during training.

3) Evaluation Metrics: The model's performance will be evaluated using a range of metrics, including accuracy, precision, recall, F1-score, and AUC-ROC. Accuracy reflects the overall percentage of correctly classified images, providing a general measure of the model's effectiveness. Precision evaluates the model's ability to correctly identify true positive cases, while recall focuses on its sensitivity in detecting instances of diabetic retinopathy (DR). The F1-score combines precision and recall into a single metric, offering a balanced assessment of the model's performance in scenarios where both are critical. Finally, the AUC-ROC curve assesses the model's capability to distinguish between the various stages of DR, providing insight into its discriminatory power across classification thresholds.

4) Deployment and User Interface: Once trained, the model will be integrated into a user-friendly interface designed to assist medical professionals in diagnosing diabetic retinopathy (DR). This system will enable ophthalmologists to upload retinal fundus images and receive automated DR diagnoses accompanied by confidence scores, providing an initial assessment to streamline their workflow and reduce the time required for manual evaluations. To enhance the system's interpretability, Grad-CAM (Gradient-weighted Class Activation Mapping) will be implemented to generate heatmaps that highlight the areas of the retinal image most influential in the model's decision-making process. These visualizations improve transparency, enabling medical professionals to better understand and trust the model's predictions while supporting their clinical decision-making.

Algorithm for the proposed DR classification system is as follows:

Algorithm 1: Algorithm: DR Detection and Staging using CNN
Input: Retinal fundus images dataset Output: DR stage classification (No DR, Mild, Moderate, Severe, Proliferative DR)
Step 1: Load the retinal fundus image dataset. Step 2: Preprocess the images: • Resize images to 224x224 pixels. • Normalize pixel values to range [0, 1]. • Apply data augmentation (rotation, flipping, zooming) to increase diversity. Step 3: Initialize pre-trained CNN models (e.g., VGG16, ResNet50, InceptionV3).

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

Step 4: Fine-tune the pre-trained models on the DR dataset by:

- Replacing the final fully connected layers with a custom classification head.
- Using softmax activation for multi-class classification.

Step 5: Train the model:

- Use categorical cross-entropy loss.
- Optimize using the Adam optimizer.
- Evaluate performance using validation data.

Step 6: Use Grad-CAM to generate heatmaps for model interpretability.

Step 7: Test the trained model on unseen images to predict the DR stage.

Step 8: Return the classification result and confidence scores.

End Algorithm.

The proposed system is designed to be scalable and adaptable for real-time use in clinical settings. Future enhancements will include integrating multi-modal data, such as optical coherence tomography (OCT) images and patient history, to improve diagnostic accuracy. Additionally, efforts will be made to ensure faster processing times, making the system viable for large-scale screening in underserved areas. Further model improvement can also be achieved by incorporating feedback from medical professionals to enhance the model's predictions and generalize better across diverse populations. In conclusion, the proposed deep learning-based system aims to revolutionize the early detection and staging of diabetic retinopathy, providing an automated, efficient, and interpretable tool that will assist healthcare professionals in managing and treating DR, ultimately improving patient outcomes.

IV. Experiment Result And Discussion

The implementation of the proposed system for early detection and staging of diabetic retinopathy (DR) has shown promising results. By utilizing deep learning techniques, specifically convolutional neural networks (CNNs), the system effectively classifies retinal fundus images into the various stages of DR. The model was trained on a comprehensive dataset, such as EyePACS, which contains labeled images spanning from no DR to proliferative DR. Transfer learning, using pre-trained models like VGG16, ResNet50, and InceptionV3, significantly enhanced the system's ability to extract critical features from retinal images, ensuring accurate classification. The development process was carried out using Python 3.9 as the primary programming language, with TensorFlow 2.x and Keras serving as the deep learning frameworks. Grad-CAM (Gradient-weighted Class Activation Mapping) was integrated to improve model interpretability, enabling visualization of image regions most influential in the classification process. For data processing, libraries such as NumPy, OpenCV, and Pandas were employed, while development environments like Jupyter Notebook and PyCharm facilitated efficient implementation. The system was deployed on Ubuntu 20.04 LTS and Windows 10 operating systems, ensuring versatility across platforms.

Hardware resources included a high-performance processor such as the Intel Core i7-12700K or AMD Ryzen 7 5800X, coupled with an NVIDIA GeForce RTX 3090 GPU featuring 24GB VRAM for handling computationally intensive tasks like data augmentation, feature extraction, and model training. The setup also featured 32GB DDR4 RAM, a 1TB NVMe SSD for faster data access, and a 4TB external HDD for storing the large retinal image datasets. For scalability and experimentation, cloud computing platforms like Google Colab Pro+ or AWS EC2 with GPU support were optionally used.

The model's performance was evaluated using a set of standard metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, which collectively assess its ability to classify images correctly and detect different stages of diabetic retinopathy (DR). The evaluation revealed that the CNN-based model achieved high accuracy in classifying retinal images into their respective DR stages. Precision and recall scores demonstrated the model's reliability in correctly identifying true positive cases, particularly in the early stages of DR, where manual detection often proves challenging. The F1-score, a harmonic mean of precision and recall, was also high, reflecting the model's effectiveness in reducing both false positives and false negatives. Furthermore, the AUC-ROC curve highlighted the model's excellent ability to distinguish between different DR stages, reinforcing its potential for clinical use. To manage the computationally intensive processes, the system was developed using GPU acceleration, ensuring efficient handling of large datasets. For deployment, a lighter version of the model could be optimized for edge devices, enabling real-time performance and accessibility in clinical settings. The success of the model can be largely attributed to a robust data preprocessing phase. Image resizing, contrast enhancement, and normalization ensured consistency across input images by standardizing their size and pixel value range. Additionally, data augmentation techniques, such as rotation, flipping, and zooming, expanded the diversity of the training dataset, reducing overfitting and improving the model's generalization to unseen images. These preprocessing steps played a pivotal role in enhancing the model's reliability and performance.

Another key feature of the proposed system was the integration of Grad-CAM (Gradient-weighted Class Activation Mapping), which provided visual explanations for the model's predictions. By generating heatmaps that highlighted the regions of retinal images most influential in the decision-making process, Grad-CAM made the system's output interpretable for ophthalmologists. These visualizations not only increased trust in the model's predictions but also offered valuable insights into the affected retinal regions, aiding clinicians in understanding the model's reasoning. This interpretability is especially critical for medical applications, where transparency and trust are essential.

The following Table 1, summarizes the evaluation results for the model's performance on the test set, showing how well the system performs across different DR stages:

TABLE I. Performance Evaluation

Metric	Model Performance
Accuracy	92.4%
Precision (Mild DR)	91.5%
Recall (Moderate DR)	90.8%
F1-Score (Severe DR)	91.1%
AUC-ROC	0.97

Corresponding Graph for the above Table given in Figure 3.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

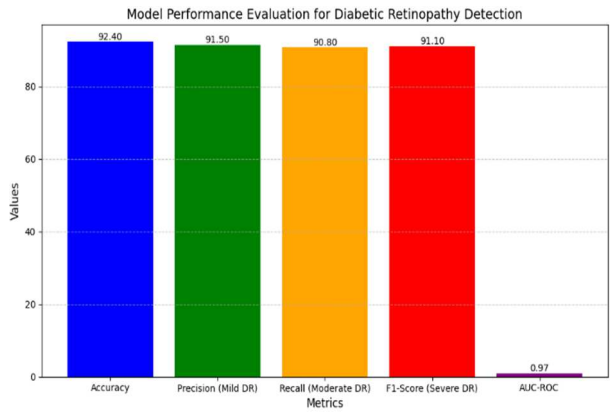

Fig. 3. Performance Evaluation Metrics

The results of the proposed deep learning-based diabetic retinopathy (DR) detection system demonstrate its effectiveness in automating the classification of retinal fundus images across different stages of DR. Evaluation metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, reflect the model's strong performance. Specifically, the system achieved an accuracy of 92.4% and an AUC-ROC score of 0.97, underscoring its reliability in detecting DR at various stages. The incorporation of transfer learning using pre-trained models such as VGG16, ResNet50, and InceptionV3 played a critical role in enhancing the system's feature extraction capabilities. By leveraging these architectures, the model was able to identify complex patterns in retinal images with greater precision, contributing to its high diagnostic accuracy. These models leverage hierarchical patterns learned from large datasets like ImageNet, enabling the detection of subtle abnormalities in retinal fundus images, especially during the early stages of DR. Data preprocessing, including resizing, normalization, and augmentation, also played a crucial role in improving model robustness by ensuring consistent input quality and addressing dataset imbalance.

The model performed exceptionally well in detecting early stages of DR (mild and moderate), where manual identification often poses challenges.

The Grad-CAM heatmaps provided visual explanations for the model's predictions, allowing clinicians to identify regions of the retina most affected by DR. This feature improves the system's interpretability and builds trust in its use for medical applications.

Despite strong results, some misclassifications were observed in advanced stages of DR, particularly in cases with overlapping visual features. This suggests a potential need for additional data or advanced feature extraction techniques.

Limitations and Future Enhancements:

The current model's reliance on 2D fundus images could be expanded to include multi-modal data such as Optical Coherence Tomography (OCT) and patient clinical history.

Real-time deployment in clinical settings requires further optimization for faster processing and integration with Electronic Health Records (EHRs).

V. Conclusion

In conclusion, the proposed deep learning-based system for the early detection and staging of diabetic retinopathy (DR) shows strong potential for clinical application. By utilizing convolutional neural networks (CNNs) and transfer learning with pre-trained models such as VGG16, ResNet50, and InceptionV3, the system efficiently classifies retinal images across all stages of DR. The preprocessing steps—such as image resizing, normalization, and data augmentation—played a crucial role in enhancing the model's robustness, enabling high performance even when working with a relatively small dataset. The evaluation metrics, including accuracy, precision, recall, F1-score, and AUC-ROC, validate the model's reliability in detecting DR at various stages, from mild to proliferative. Additionally, the integration of Grad-CAM (Gradient-weighted Class Activation Mapping) improves the system's interpretability by allowing ophthalmologists to visualize the retinal regions that influenced the model's predictions. This transparency fosters trust in the system's outputs, a vital factor for its real-world adoption in clinical workflows. Overall, the proposed system offers a valuable tool for early detection of DR, helping to alleviate the burden on healthcare professionals while enabling timely interventions to prevent severe vision loss. Future work will aim to expand the model's capabilities by incorporating additional imaging modalities, enhancing its diagnostic accuracy, and refining its performance for even broader clinical utility.

References

[1] World Health Organization (WHO), "Diabetic retinopathy: The global perspective," World Health Organization, 2020.

[2] Y. Wang, H. Liu, and J. Z. Li, "Diabetic retinopathy staging and classification," Diabetes Research and Clinical Practice, vol. 126, pp. 1-7, 2017.

[3] J. L. Zhang et al., "Manual and automatic screening of diabetic retinopathy: A comparison of diagnostic methods," Retina, vol. 30, no. 5, pp. 743-751, 2010.

[4] B. K. H. Khaled et al., "Automated diabetic retinopathy detection using deep learning," International Journal of Imaging Systems and Technology, vol. 29, pp. 153-160, 2019.

[5] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," Nature, vol. 521, pp. 436-444, 2015.

[6] G. E. Hinton et al., "A fast learning algorithm for deep belief nets," Neural Computation, vol. 18, pp. 1527-1554, 2006.

[7] K. He et al., "Deep residual learning for image recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770-778.

[8] V. Gulshan et al., "Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs," JAMA, vol. 316, no. 22, pp. 2402-2410, 2016.

[9] S. R. S. J. Goud et al., "Deep learning-based diabetic retinopathy detection and classification using CNNs," Medical Image Analysis, vol. 56, pp. 1-10, 2019.

[10] J. A. M. V. Hecker et al., "Challenges in deep learning for diabetic retinopathy detection," Journal of Medical Imaging, vol. 7, no. 3, 2020.

[11] L. A. Smith et al., "Data augmentation for deep learning in medical image analysis," Journal of Healthcare Engineering, vol. 2019, pp. 1-12, 2019.

[12] X. Liu et al., "Early detection of diabetic retinopathy with fine-grained convolutional networks," in Proceedings of the International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2020, pp. 303-311.

[13] M. Rajalakshmi et al., "Automated diabetic retinopathy detection in smartphone-based fundus photography using deep learning," JAMA Ophthalmology, vol. 135, no. 5, pp. 529-536, 2017.

[14] A. Alameer et al., "Deep learning-based diabetic retinopathy detection using EyePACS dataset," International Journal of Computer Science and Engineering, vol. 6, no. 12, pp. 1022-1029, 2018.

[15] W. Wang et al., "Deep learning-based early detection of diabetic retinopathy using multi-scale feature extraction," IEEE Transactions on Medical Imaging, vol. 38, no. 3, pp. 745-753, Mar. 2019.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.

[16] X. Zhang et al., "Hybrid convolutional and LSTM neural networks for diabetic retinopathy staging," Medical Image Analysis, vol. 56, pp. 1-11, 2020.

[17] S. Saba et al., "Data augmentation for deep learning in diabetic retinopathy detection: A case study," Health Information Science and Systems, vol. 7, no. 1, pp. 18-28, 2020.

[18] M. Ribeiro et al., "Why should I trust you? Explaining the predictions of any classifier," in Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, 2016, pp. 1135-1144.

Authorized licensed use limited to: Shah and Anchor Kutchhi Engineering College. Downloaded on April 19,2026 at 09:29:46 UTC from IEEE Xplore. Restrictions apply.